

## SAFETY DATA SHEET

Revision Date 01-April-2024

Revision Number 3

### 1. Identification

**Product Name** Holmium(III) isopropoxide, 5% w/v in toluene/isopropanol

**Cat No. :** 36579

**Synonyms** No information available

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

**Importer/Distributor**  
Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                                   |              |
|-------------------------------------------------------------------|--------------|
| <b>Flammable liquids</b>                                          | Category 2   |
| <b>Skin Corrosion/Irritation</b>                                  | Category 1 B |
| <b>Serious Eye Damage/Eye Irritation</b>                          | Category 1   |
| <b>Reproductive Toxicity</b>                                      | Category 2   |
| <b>Specific target organ toxicity (single exposure)</b>           | Category 3   |
| Target Organs - Respiratory system, Central nervous system (CNS). |              |
| <b>Specific target organ toxicity - (repeated exposure)</b>       | Category 2   |
| Target Organs - Neurological effects, Eyes, Ears.                 |              |
| <b>Aspiration Toxicity</b>                                        | Category 1   |

#### Label Elements

**Signal Word**  
Danger

### **Hazard Statements**

Highly flammable liquid and vapor  
May be fatal if swallowed and enters airways  
Causes severe skin burns and eye damage  
May cause respiratory irritation  
May cause drowsiness and dizziness  
Suspected of damaging the unborn child  
May cause damage to organs through prolonged or repeated exposure



### **Precautionary Statements**

#### **Prevention**

Obtain special instructions before use  
Do not handle until all safety precautions have been read and understood  
Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
Keep container tightly closed  
Ground/bond container and receiving equipment  
Do not breathe dust/fumes/gas/mist/vapours/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Use only outdoors or in a well-ventilated area  
Wear protective gloves/protective clothing/eye protection/face protection  
Use non-sparking tools  
Take action to prevent static discharges

#### **Response**

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower  
IF INHALED: Remove person to fresh air and keep comfortable for breathing  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Immediately call a POISON CENTER/doctor  
Rinse mouth  
Do NOT induce vomiting  
Wash contaminated clothing before reuse  
In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

#### **Storage**

Store locked up  
Store in a well-ventilated place. Keep container tightly closed

#### **Disposal**

Dispose of contents/container to an approved waste disposal plant

## **3. Composition/Information on Ingredients**

| <b>Component</b>          | <b>CAS-No</b> | <b>Weight %</b> |
|---------------------------|---------------|-----------------|
| Toluene                   | 108-88-3      | 47.50           |
| Isopropyl alcohol         | 67-63-0       | 47.50           |
| Holmium(III) isopropoxide | 30429-72-2    | 5.00            |

## **4. First-aid measures**

### **General Advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | required.                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Eye Contact</b>                     | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                                                                                                             |
| <b>Skin Contact</b>                    | Wash off immediately with plenty of water for at least 15 minutes. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Call a physician immediately.                                                                                                                                                                                                                                                       |
| <b>Inhalation</b>                      | If not breathing, give artificial respiration. Remove from exposure, lie down. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Call a physician immediately. Risk of serious damage to the lungs (by aspiration).                                                           |
| <b>Ingestion</b>                       | Do NOT induce vomiting. Clean mouth with water. Never give anything by mouth to an unconscious person. Call a physician immediately. Call a physician or poison control center immediately. If vomiting occurs naturally, have victim lean forward.                                                                                                                                                                                           |
| <b>Most important symptoms/effects</b> | Causes burns by all exposure routes. Difficulty in breathing. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                                                                                                                                                                                                                                                                         |

## 5. Fire-fighting measures

|                                         |                                                                                                                     |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | CO <sub>2</sub> , dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                                                            |
| <b>Flash Point</b>                      | 4 °C / 39.2 °F                                                                                                      |
| <b>Method -</b>                         | No information available                                                                                            |
| <b>Autoignition Temperature</b>         | No information available                                                                                            |
| <b>Explosion Limits</b>                 |                                                                                                                     |
| <b>Upper</b>                            | No data available                                                                                                   |
| <b>Lower</b>                            | No data available                                                                                                   |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                            |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                            |

### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Holmium oxide.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### NFPA

|               |                     |                    |                         |
|---------------|---------------------|--------------------|-------------------------|
| <b>Health</b> | <b>Flammability</b> | <b>Instability</b> | <b>Physical hazards</b> |
| 3             | 3                   | 0                  | -                       |

## 6. Accidental release measures

|                                             |                                                                                                                                                                                                                                                      |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Remove all sources of ignition. Take precautionary measures against static discharges. |
| <b>Environmental Precautions</b>            | Do not flush into surface water or sanitary sewer system. Should not be released into the environment. Do not allow material to contaminate ground water system.                                                                                     |
| <b>Methods for Containment and Clean Up</b> | Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.                                                                        |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges. |
| <b>Storage.</b> | Corrosives area. Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame.                                                                                                                                                                                                                                                                                                                                                                                  |

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component         | Alberta                                                                                    | British Columbia              | Ontario TWA/EV                | Quebec                        | ACGIH TLV                     | OSHA PEL                                                                                                                                                                          | NIOSH                                                                                                         |
|-------------------|--------------------------------------------------------------------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Toluene           | TWA: 50 ppm<br>TWA: 188 mg/m <sup>3</sup><br>Skin                                          | TWA: 20 ppm                   | TWA: 20 ppm                   | TWA: 20 ppm                   | TWA: 20 ppm                   | (Vacated) TWA: 100 ppm<br>(Vacated) TWA: 375 mg/m <sup>3</sup><br>Ceiling: 300 ppm<br>(Vacated) STEL: 150 ppm<br>(Vacated) STEL: 560 mg/m <sup>3</sup><br>TWA: 200 ppm            | IDLH: 500 ppm<br>TWA: 100 ppm<br>TWA: 375 mg/m <sup>3</sup><br>STEL: 150 ppm<br>STEL: 560 mg/m <sup>3</sup>   |
| Isopropyl alcohol | TWA: 200 ppm<br>TWA: 492 mg/m <sup>3</sup><br>STEL: 400 ppm<br>STEL: 984 mg/m <sup>3</sup> | TWA: 200 ppm<br>STEL: 400 ppm | TWA: 200 ppm<br>STEL: 400 ppm | TWA: 200 ppm<br>STEL: 400 ppm | TWA: 200 ppm<br>STEL: 400 ppm | (Vacated) TWA: 400 ppm<br>(Vacated) TWA: 980 mg/m <sup>3</sup><br>(Vacated) STEL: 500 ppm<br>(Vacated) STEL: 1225 mg/m <sup>3</sup><br>TWA: 400 ppm<br>TWA: 980 mg/m <sup>3</sup> | IDLH: 2000 ppm<br>TWA: 400 ppm<br>TWA: 980 mg/m <sup>3</sup><br>STEL: 500 ppm<br>STEL: 1225 mg/m <sup>3</sup> |

### Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

### Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting/equipment. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control

hazardous materials at source

### Personal protective equipment

**Eye Protection** Goggles  
**Hand Protection** Wear appropriate protective gloves and clothing to prevent skin exposure.

| Glove material              | Breakthrough time                 | Glove thickness | Glove comments         |
|-----------------------------|-----------------------------------|-----------------|------------------------|
| Nitrile rubber<br>Viton (R) | See manufacturers recommendations | -               | Splash protection only |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Organic gases and vapours filter low boiling organic solvent Type AX Brown conforming to EN371 or Type A Brown conforming to EN14387

When RPE is used a face piece Fit Test should be conducted

### Environmental exposure controls

Prevent product from entering drains. Do not allow material to contaminate ground water system.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                                        |                          |
|----------------------------------------|--------------------------|
| Physical State                         | Liquid                   |
| Appearance                             | Pink                     |
| Odor                                   | No information available |
| Odor Threshold                         | No information available |
| pH                                     | No information available |
| Melting Point/Range                    | No data available        |
| Boiling Point/Range                    | No information available |
| Flash Point                            | 4 °C / 39.2 °F           |
| Evaporation Rate                       | No information available |
| Flammability (solid,gas)               | Not applicable           |
| Flammability or explosive limits       |                          |
| Upper                                  | No data available        |
| Lower                                  | No data available        |
| Vapor Pressure                         | No information available |
| Vapor Density                          | No information available |
| Specific Gravity                       | No information available |
| Solubility                             | No information available |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition Temperature              | No information available |
| Viscosity                              | No information available |
| Molecular Formula                      | C9 H21 HoO3              |
| Molecular Weight                       | 342.19                   |

## 10. Stability and reactivity

|                                         |                                                                        |
|-----------------------------------------|------------------------------------------------------------------------|
| <b>Reactive Hazard</b>                  | None known, based on information available                             |
| <b>Stability</b>                        | Moisture sensitive.                                                    |
| <b>Conditions to Avoid</b>              | Keep away from open flames, hot surfaces and sources of ignition.      |
| <b>Incompatible Materials</b>           | Strong oxidizing agents                                                |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Holmium oxide |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                               |
| <b>Hazardous Reactions</b>              | None under normal processing.                                          |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

##### Oral LD50

Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg.

##### Dermal LD50

Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg.

##### Vapor LC50

Based on ATE data, the classification criteria are not met. ATE > 20 mg/l.

#### Component Information

| Component         | LD50 Oral                                  | LD50 Dermal                   | LC50 Inhalation       |
|-------------------|--------------------------------------------|-------------------------------|-----------------------|
| Toluene           | > 5000 mg/kg ( Rat )                       | LD50 = 12000 mg/kg ( Rabbit ) | 26700 ppm ( Rat ) 1 h |
| Isopropyl alcohol | 5045 mg/kg ( Rat )<br>3600 mg/kg ( Mouse ) | 12800 mg/kg ( Rat )           | 72.6 mg/L ( Rat ) 4 h |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** No information available

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                 | CAS-No     | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|---------------------------|------------|------------|------------|------------|------------|------------|
| Toluene                   | 108-88-3   | Not listed | Not listed | Not listed | Not listed | Not listed |
| Isopropyl alcohol         | 67-63-0    | Not listed | Not listed | Not listed | Not listed | Not listed |
| Holmium(III) isopropoxide | 30429-72-2 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** California Proposition 65. Reproductive toxicity.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** Respiratory system Central nervous system (CNS)

**STOT - repeated exposure** Neurological effects Eyes Ears

**Aspiration hazard** No information available

**Symptoms / effects, both acute and** Inhalation of high vapor concentrations may cause symptoms like headache, dizziness,

**delayed** tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

The product contains following substances which are hazardous for the environment. Contains a substance which is: Toxic to aquatic organisms. May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

| Component         | Freshwater Algae                                                                                                           | Freshwater Fish                                                                                                                                                                                              | Microtox                                           | Water Flea                                                                                   |
|-------------------|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|----------------------------------------------------------------------------------------------|
| Toluene           | EC50: = 12.5 mg/L, 72h static (Pseudokirchneriella subcapitata)<br>EC50: > 433 mg/L, 96h (Pseudokirchneriella subcapitata) | 50-70 mg/L LC50 96 h<br>5-7 mg/L LC50 96 h<br>15-19 mg/L LC50 96 h<br>28 mg/L LC50 96 h<br>12 mg/L LC50 96 h                                                                                                 | EC50 = 19.7 mg/L 30 min                            | EC50: = 11.5 mg/L, 48h (Daphnia magna)<br>EC50: 5.46 - 9.83 mg/L, 48h Static (Daphnia magna) |
| Isopropyl alcohol | EC50: > 1000 mg/L, 72h (Desmodesmus subspicatus)<br>EC50: > 1000 mg/L, 96h (Desmodesmus subspicatus)                       | LC50: = 9640 mg/L, 96h flow-through (Pimephales promelas)<br>LC50: > 1400000 µg/L, 96h (Lepomis macrochirus)<br>LC50: = 11130 mg/L, 96h static (Pimephales promelas)<br>LC50: = 10000000 µg/L, 96h (Daphnia) | = 35390 mg/L EC50 Photobacterium phosphoreum 5 min | 13299 mg/L EC50 = 48 h<br>9714 mg/L EC50 = 24 h                                              |

**Persistence and Degradability** Immiscible with water May persist

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Is not likely mobile in the environment due its low water solubility.

| Component         | log Pow |
|-------------------|---------|
| Toluene           | 2.73    |
| Isopropyl alcohol | 0.05    |

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

| Component          | RCRA - U Series Wastes | RCRA - P Series Wastes |
|--------------------|------------------------|------------------------|
| Toluene - 108-88-3 | U220                   | -                      |

## 14. Transport information

### DOT

|                      |                                            |
|----------------------|--------------------------------------------|
| UN-No                | UN1993                                     |
| Proper Shipping Name | Flammable liquid, n.o.s.                   |
| Technical Name       | (ISOPROPANOL (ISOPROPYL ALCOHOL), TOLUENE) |
| Hazard Class         | 3                                          |
| Packing Group        | II                                         |

**TDG**

UN-No UN1993  
 Proper Shipping Name Flammable liquid, n.o.s.  
 Hazard Class 3  
 Packing Group II

**IATA**

UN-No UN1993  
 Proper Shipping Name Flammable liquid, n.o.s.  
 Hazard Class 3  
 Packing Group II

**IMDG/IMO**

UN-No UN1993  
 Proper Shipping Name Flammable liquid, n.o.s.  
 Hazard Class 3  
 Packing Group II

## 15. Regulatory information

## International Inventories

| Component                 | CAS-No     | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|---------------------------|------------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Toluene                   | 108-88-3   | X   | -    | X    | ACTIVE                                        | 203-625-9 | -      | -   |
| Isopropyl alcohol         | 67-63-0    | X   | -    | X    | ACTIVE                                        | 200-661-7 | -      | -   |
| Holmium(III) isopropoxide | 30429-72-2 | -   | -    | -    | -                                             | -         | -      | -   |

| Component                 | CAS-No     | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|---------------------------|------------|-------|----------|------|------|------|------|-------|-------|
| Toluene                   | 108-88-3   | X     | KE-33936 | X    | X    | X    | X    | X     | X     |
| Isopropyl alcohol         | 67-63-0    | X     | KE-29363 | X    | X    | X    | X    | X     | X     |
| Holmium(III) isopropoxide | 30429-72-2 | -     | -        | -    | -    | -    | -    | -     | -     |

**Legend:**

X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances**IECSC** - Chinese Inventory of Existing Chemical Substances**KECL** - Korean Existing and Evaluated Chemical Substances**ENCS** - Japanese Existing and New Chemical Substances**AICS** - Australian Inventory of Chemical Substances**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

## Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

| Component         | Canada - National Pollutant Release Inventory (NPRI)                        | Canadian Environmental Protection Agency (CEPA) - List of Toxic Substances | Canada's Chemicals Management Plan (CEPA) |
|-------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------|
| Toluene           | Part 1, Group A Substance<br>Part 5, Individual Substances Part 4 Substance |                                                                            |                                           |
| Isopropyl alcohol | Part 1, Group A Substance<br>Part 5, Individual Substances Part 4 Substance |                                                                            |                                           |

**Legend**

NPRI - National Pollutant Release Inventory

## Other International Regulations



Authorisation/Restrictions according to EU REACH

| Component         | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                            | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-------------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Toluene           | -                                                                   | Use restricted. See item 48.<br>(see link for restriction details)<br>Use restricted. See item 75.<br>(see link for restriction details) | -                                                                                                     |
| Isopropyl alcohol | -                                                                   | Use restricted. See item 75.<br>(see link for restriction details)                                                                       | -                                                                                                     |

REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component                 | CAS-No     | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|---------------------------|------------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Toluene                   | 108-88-3   | Listed         | Not applicable               | Not applicable            | Not applicable                             |
| Isopropyl alcohol         | 67-63-0    | Listed         | Not applicable               | Not applicable            | Not applicable                             |
| Holmium(III) isopropoxide | 30429-72-2 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component                 | CAS-No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|---------------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Toluene                   | 108-88-3   | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Annex I - Y42                      |
| Isopropyl alcohol         | 67-63-0    | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Annex I - Y42                      |
| Holmium(III) isopropoxide | 30429-72-2 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

Prepared By

Product Safety Department  
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www.thermofisher.com

Revision Date

01-April-2024

Print Date

01-April-2024

Revision Summary

New emergency telephone response service provider.

Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**